C – Functions

1a) $D_f = (1,\infty)$ since $x-1>0 \Rightarrow x>1$

$f(5) = \frac{1}{\sqrt{4}} = \frac{1}{2}$

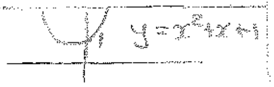
b) $D_f = \mathbb{R}\setminus\{4\}$ since $|x-4|\neq 0 \Rightarrow |x-4|>0 \Rightarrow x>4$ or $x<4 \Rightarrow x\neq 4$

$f(1) = \frac{3}{|-3|} = 1$

(c) $D_f = \mathbb{R}$ f is a polynomial

$f(3) = 3^4+3^3+1 = 81+27+1 = 109$

(d) $f(x) = \frac{1}{x^2+x+1}$



$D_f = \mathbb{R}$ since $y = x^2+x+1>0$

$f(0) = \frac{1}{1} = 1$

e) $\sin x = 0$

$\Rightarrow x = 0+2k\pi$ or $x = \pi+2k\pi$ $k\in\mathbb{Z}$

$\Rightarrow x = \pi k, k\in\mathbb{Z}$

$\therefore D_f = \{x \mid x\neq \pi k, k\in\mathbb{Z}\}$

$f(5\pi/2) = \sin 5\pi/2 = \sin \pi/2 = 1$ $5\pi/2 = 2\times2\pi+\pi/2 = \pi/2$

f) $D_f = \mathbb{R}\setminus\{0\}$

$f(\ln 3) = e^{\frac{1}{\ln 3}}$

g) $D_f = (0,1)\cup(1,\infty)$ since $\sqrt{x}>0$ and $\sqrt{x}-1\neq 0 \Rightarrow x\neq 1$

$f(2) = \frac{\sqrt{2}+1}{\sqrt{2}-1}$

h) $y = \tan x$ is not defined for $x = \pi/2+2k\pi$, $k\in\mathbb{Z}$

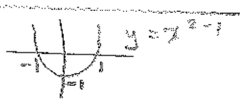
$\therefore D_f = \{x \mid x\neq \pi/2+2k\pi, k\in\mathbb{Z}\}$

$f(-\pi/4) = \tan(-\pi/4) = -1$

(i) $D_f = \mathbb{R}\setminus\{0\}$ since $x\neq 0$

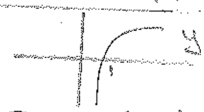
$f(-1) = \frac{|-1|}{-1} = -1$

(j) $x^2-1\neq 0 \Rightarrow x\neq\pm 1$



$D_f = \mathbb{R}\setminus\{-1,1\}$ $f(0) = \frac{1}{-1} = -1$

(k)



$y=\ln x \Rightarrow x>0$ and $x-1\neq 0 \Rightarrow x\neq 1$

$D_f = (0,1)\cup(1,\infty)$

$f(e) = \frac{\ln e}{e-1} = \frac{1}{e-1}$

(l) $f(x) = \ln(1+e^x)$

$\Rightarrow \underbrace{1+e^x}_{\text{always}>0}>0 \Rightarrow x\in\mathbb{R}$

$f(0) = \ln(1+e^0) = \ln 2 \approx 0.7$

2(a) $(f+g)(x) = 2x+5+x^2,\ x\in\mathbb{R}$

$(f-g)(x) = 2x+5-x^2,\ x\in\mathbb{R}$

$(fg)(x) = (2x+5)(x^2),\ x\in\mathbb{R}$

$\left(\frac{f}{g}\right)(x) = \frac{2x+5}{x^2},\ x\neq 0$

$(f\circ g)(x) = f(x^2) = 2x^2+5,\ x\in\mathbb{R}$

$(g\circ f)(x) = g(2x+5) = (2x+5)^2,\ x\in\mathbb{R}$

(b) $(f+g)(x) = x^2+\frac{1}{x-1},\ x\neq 1$

$(f-g)(x) = x^2-\frac{1}{x-1},\ x\neq 1$

$(fg)(x) = \frac{x^2}{x-1},\ x\neq 1$

$\left(\frac{f}{g}\right)(x) = \frac{x^2}{\frac{1}{x-1}} = x^2(x-1),\ x\neq 1$

$(f\circ g)(x) = f\left(\frac{1}{x-1}\right) = \frac{1}{(x-1)^2},\ x\neq 1$

$(g\circ f)(x) = g(x^2) = \frac{1}{x^2-1},\ x\neq\pm 1$

(c) $(f+g)(x) = \underbrace{e^x}_{x\in\mathbb{R}}+\underbrace{\ln x}_{x>0},\ x>0$

$(f-g)(x) = e^x-\ln x,\ x>0$

$(fg)(x) = e^x\ln x,\ x>0$

$\left(\frac{f}{g}\right)(x) = \frac{e^x}{\ln x},\ x>0$ and $\ln x\neq 0$

$\Rightarrow x>0$ and $x\neq 1$

$(f\circ g)(x) = f(\underbrace{\ln x}_{D_g=(0,\infty)}) = e^{\ln x} = x,\ x>0$

$(g\circ f)(x) = g(\underbrace{e^x}_{D_f=\mathbb{R}}) = \ln e^x = x,\ x\in\mathbb{R}$

(d) $(f+g)(x) = \cos x+\frac{1}{x},\ x\neq 0$

$(f-g)(x) = \cos x-\frac{1}{x},\ x\neq 0$

$(fg)(x) = \frac{\cos x}{x},\ x\neq 0$

$\left(\frac{f}{g}\right)(x) = \frac{\cos x}{\frac{1}{x}} = x\cos x,\ x\neq 0$

$(f\circ g)(x) = f(\underbrace{\tfrac{1}{x}}_{D_g=\mathbb{R}\setminus\{0\}}) = \cos\left(\frac{1}{x}\right),\ x\neq 0$

$(g\circ f)(x) = g(\cos x) = \frac{1}{\cos x}$

$x\neq \pi/2+2k\pi,\ k\in\mathbb{Z}$

(2)

(e) $(f+g)(x) = \sqrt{x} + x - 1 \quad , x \geq 0$

$(f-g)(x) = \sqrt{x} - (x-1) \quad , x \geq 0$

$(fg)(x) = \sqrt{x}(x-1) \quad , x \geq 0$

$= x^{3/2} - x^{1/2} = \sqrt{x^3} - \sqrt{x}$

$\left(\frac{f}{g}\right)(x) = \frac{\sqrt{x}}{x-1} \quad , x \geq 0 \text{ and } x \neq 1$

$(f \circ g)(x) = \sqrt{x-1} \quad , x \geq 1$

$(g \circ f)(x) = (\sqrt{x}) - 1 \quad , x \geq 0$

(f) $(f+g)(x) = \sqrt{x} + 2^x \quad , x \geq 0$

$(f-g)(x) = \sqrt{x} - 2^x \quad , x \geq 0$

$(fg)(x) = (\sqrt{x})(2^x) \quad , x \geq 0$

$\left(\frac{f}{g}\right)(x) = \frac{\sqrt{x}}{2^x} \quad , x \geq 0$

$(f \circ g)(x) = f(2^x) = \sqrt{2^x} = (2^x)^{1/2}$

$= 2^{x/2} \quad , x \in \mathbb{R}$

$(g \circ f)(x) = g(\sqrt{x}) = 2^{\sqrt{x}} \quad , x \geq 0$